



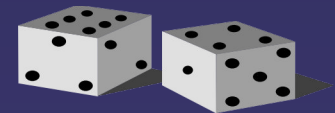
Random Variables And Probability Distributions

DEPARTMENT OF APPLIED SCIENCES
PARUL UNIVERSITY



Topics

- Probability Distributions
- Expected Value and Variance of a RV



Topics

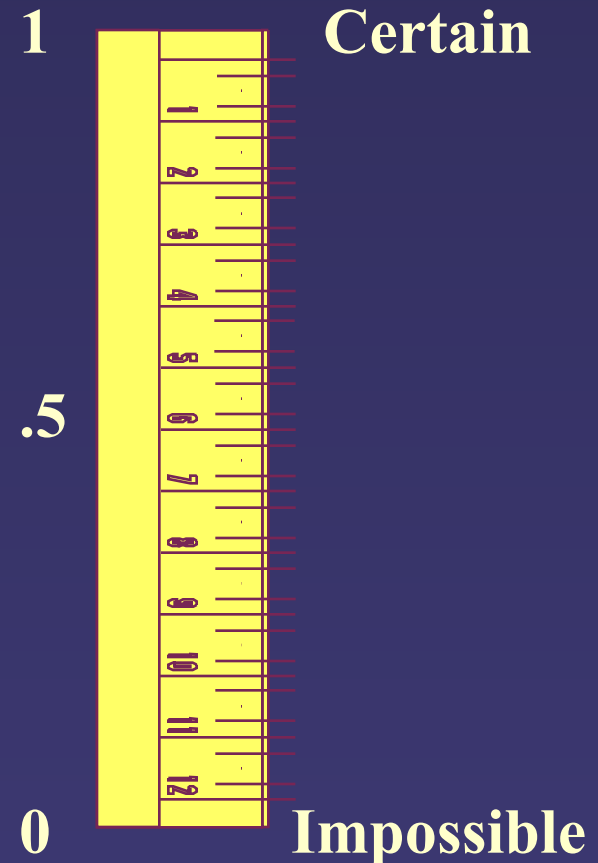
- **Discrete Probability Distributions**
 - Bernoulli and Binomial Distributions
 - Poisson Distributions
- **Continuous Probability Distributions**
 - Uniform
 - Normal

Topics

- **Random Sampling and Probability Distributions**
 - **Random Numbers**
 - **Sampling from Probability Distributions**

Probability

- Probability is the likelihood that the event will occur.
- Two Conditions:
 - Value is **between 0 and 1**.
 - **Sum** of the probabilities of all events must be **1**.

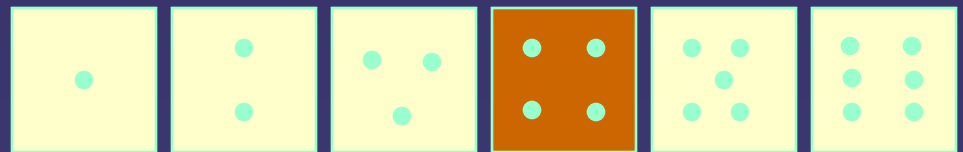


Random Variable

- A numerical description of the outcome of an experiment

Example: Discrete RV: countable # of outcomes

Throw a die twice: Count the number of times 4 comes up (0, 1, or 2 times)



Discrete Random Variable

- **Discrete Random Variable:**

- **Obtained by Counting (0, 1, 2, 3, etc.)**
- **Usually finite by number of different values**

e.g.

Toss a coin 5 times. Count the number of tails. (0, 1, 2, 3, 4, or 5 times)

Random Variable

- A numerical description of the outcome of an experiment

Example: Continuous RV:

- The Value of the DJIA
- Time to repair a failed machine
- Random Variable: Given by Capital Letters X & Y.
- Specific Values Given by lower case

Probability Distribution

Characterization of the possible values that a RV may assume along with the probability of assuming these values.

Discrete Probability Distribution

- List of all possible $[x_i, p(x_i)]$ pairs
 - x_i = value of random variable
 - $P(x_i)$ = probability associated with value
- Mutually exclusive (nothing in common)
- Collectively exhaustive (nothing left out)

$$0 \leq p(x_i) \leq 1$$


$$P(x_i) = 1$$

Weekly Demand of a Slow-Moving Product

Probability Mass Function

Demand, x	Probability, $p(x)$
0	0.1
1	0.2
2	0.4
3	0.3
4 or more	0

Weekly Demand of a Slow-Moving Product

A Cumulative Distribution Function:

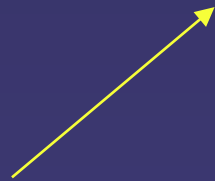
Probability that RV assume a value $\leq$ a given value, x

Demand, x	Cumulative Probability, $P(x)$
0	0.1
1	0.3
2	0.7
3	1

Discrete Probability Distribution Example

Event: Toss 2 Coins.

Count # Tails.



Values

0

1

2

Probability distribution

probability

$1/4 = .25$

$2/4 = .50$

$1/4 = .25$

Discrete Random Variable Summary Measures

Expected value (The mean)

Weighted average of the probability distribution

$$\mu = E(X) = \sum x_i p(x_i)$$

In slow-moving product demand example,
the expected value is :

$$E(X) = 0 \times 0.1 + 1 \times .2 + 2 \times .4 + 3 \times .3 = 1.9$$

The average demand on the long run is 1.9

Discrete Random Variable Summary Measures

Variance

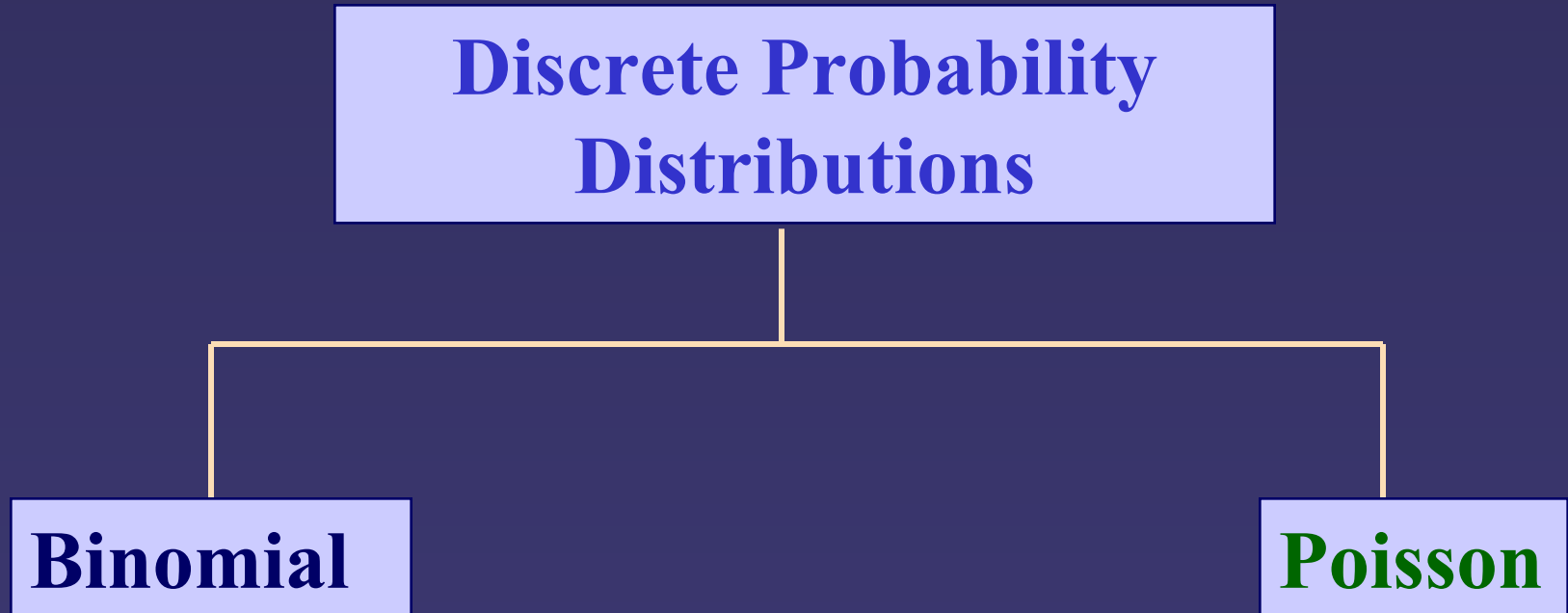
Weighted average squared deviation about mean

$$\text{Var}[X] = \sum (x_i - E(X))^2 p(x_i)$$

For the Product demand example, the variance is:

$$\begin{aligned}\text{Var}[X] &= \sum (x_i - E(X))^2 p(x_i) = (0 - 1.9)^2(.1) + (1 - 1.9)^2(.2) + \\ &\quad (2 - 1.9)^2(.4) + (3 - 1.9)^2(.3) = .89\end{aligned}$$

Important Discrete Probability Distribution Models



Binomial Distribution

- **'N' identical trials**
 - **Example: 15 tosses of a coin, 10 light bulbs taken from a warehouse**
- **2 mutually exclusive outcomes on each trial**
 - **Example: Heads or tails in each toss of a coin, defective or not defective light bulbs**

Binomial Distributions

- **Constant Probability for each Trial**
 - **Example:** Probability of getting a tail is the same each time we toss the coin and each light bulb has the same probability of being defective
- **2 Sampling Methods:**
 - **Infinite Population Without Replacement**
 - **Finite Population With Replacement**
- **Trials are Independent:**
 - **The Outcome of One Trial Does Not Affect the Outcome of Another**

Binomial Probability Distribution Function

$$P(X) = \frac{n!}{X!(n-X)!} p^X (1-p)^{n-X}$$

$P(X)$ = probability that X successes given a knowledge of n and p

X = number of 'successes' in sample, ($X = 0, 1, 2, \dots, n$)

p = probability of each 'success'

n = sample size

Tails in 2 Tosses of Coin

<u>X</u>	<u>$P(X)$</u>
0	$1/4 = .25$
1	$2/4 = .50$
2	$1/4 = .25$

Binomial Distribution

Characteristics

Mean

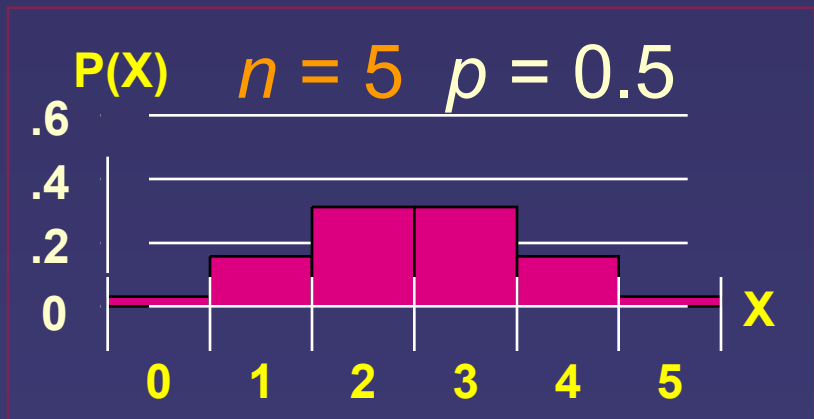
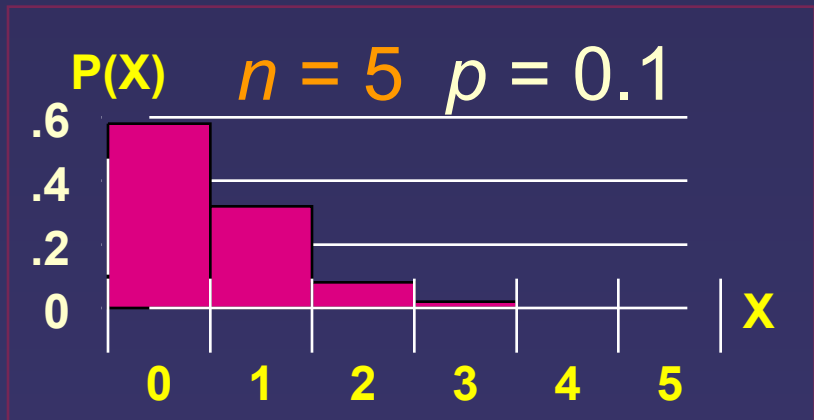
! $\bullet E(X) \bullet np$

e.g. ! $= 5 (.1) = .5$

Standard Deviation

? $\bullet \sqrt{np(1-p)}$

e.g. ? $= \sqrt{5(.5)(1-.5)}$
 $= 1.118$



Binomial Probabilities

n 10
p 0.8

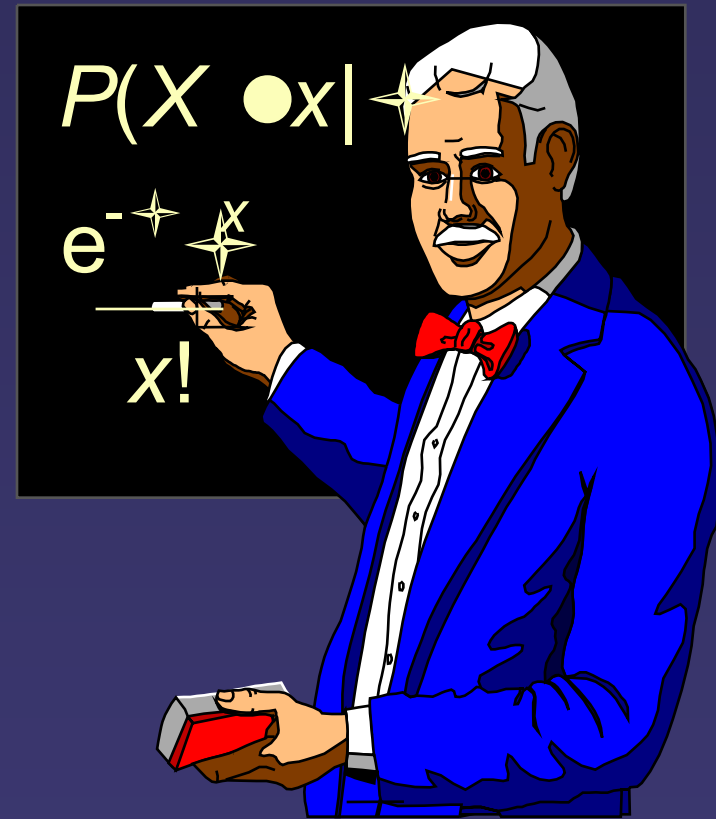
**Computing Binomial
Probabilities using Excel
Function BINOMDIST**

x	p(x)
0	0.000000
1	0.000004
2	0.000074
3	0.000786
4	0.005505
5	0.026424
6	0.088080
7	0.201327
8	0.301990
9	0.268435
10	0.107374

Poisson Distribution

Poisson process:

- Discrete events in an 'interval'
 - The probability of **one success** in an interval is **stable**
 - The probability of **more than one success** in this interval is **0**
- Probability of success is **Independent** from interval to Interval



Examples of Poisson Distribution



Poisson Distribution Function

$$P(X) = e^{-\lambda} \frac{\lambda^X}{X!}$$

$P(X)$ = probability of X successes given λ

λ = expected (mean) number of 'successes'

e = 2.71828 (base of natural logs)

X = number of 'successes' per unit

e.g. Find the probability of 4 customers arriving in 3 minutes when the mean is 3.6

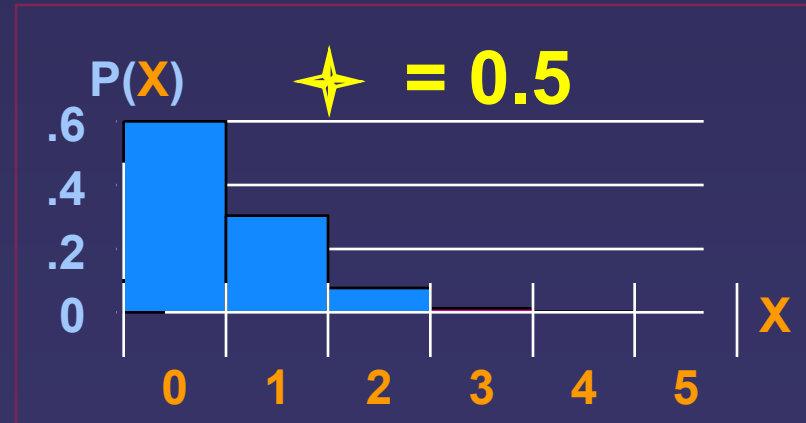
$$P(X) = e^{-3.6} \frac{3.6^4}{4!} = .1912$$

Poisson Distribution Characteristics

Mean

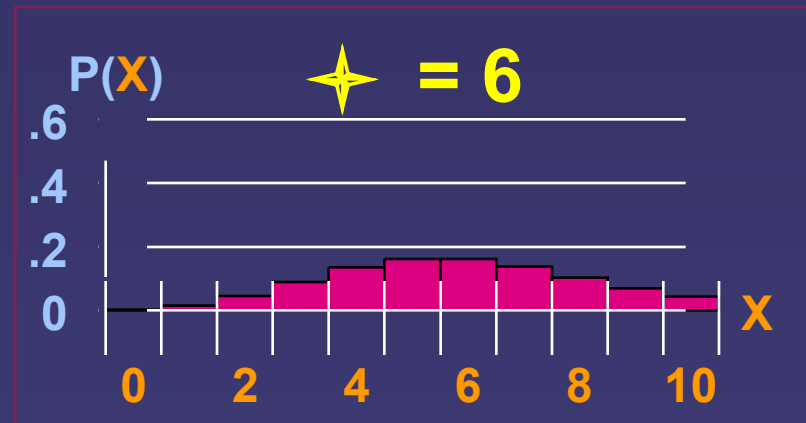
! ● $E(X)$ ● ✨

● $\sum_{i=1}^N X_i P(X_i)$



Standard Deviation

? ● $\sqrt{\lambda}$ ✨



Poisson Distribution

Mean

12

**Computing Poisson
Probabilities using Excel
Function POISSON**

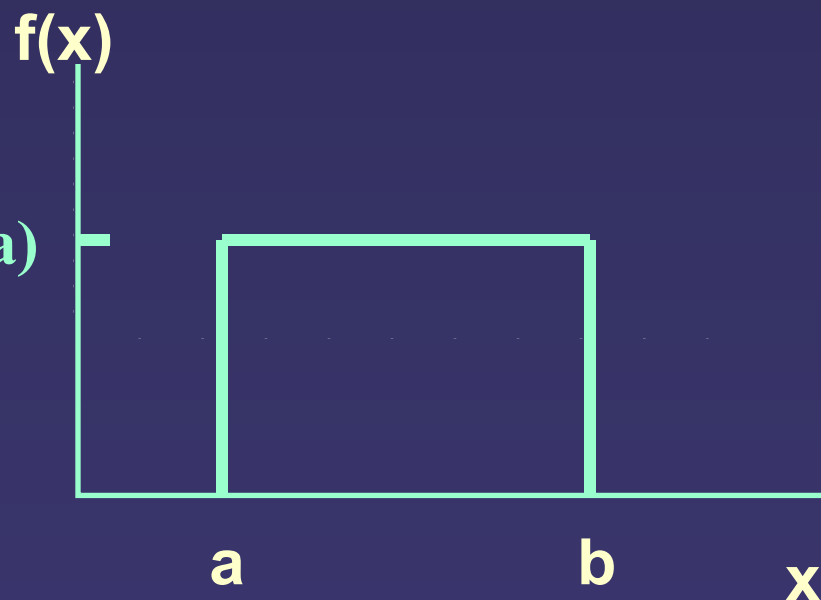
x	p(x)
1	0.00007
2	0.00044
3	0.00177
4	0.00531
5	0.01274
6	0.02548
7	0.04368
8	0.06552
9	0.08736
10	0.10484
11	0.11437
12	0.11437

Continuous Probability Distributions

- Uniform
- Normal

The Uniform Distribution

- Equally Likely chances of occurrences of RV values between a maximum and a minimum
- Mean = $(b+a)/2$
- Variance = $(b-a)^2/12$
- 'a' is a location parameter
- 'b-a' is a scale parameter
- no shape parameter



The Uniform Distribution

Probability Density Function

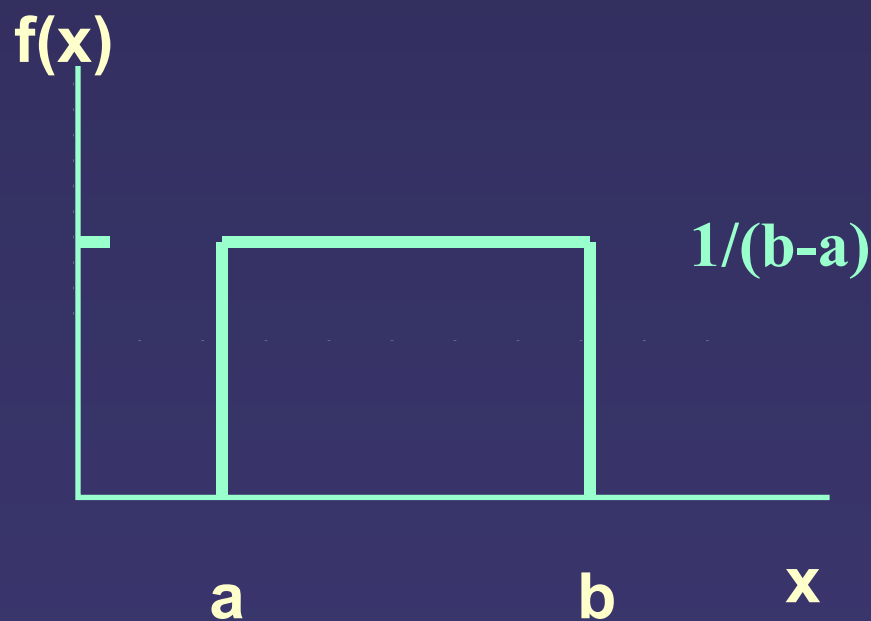
$$f(x) = \frac{1}{b - a} \quad \text{if } a \leq x \leq b$$

Distribution Function

$$F(x) = 0 \quad \text{if } x < a$$

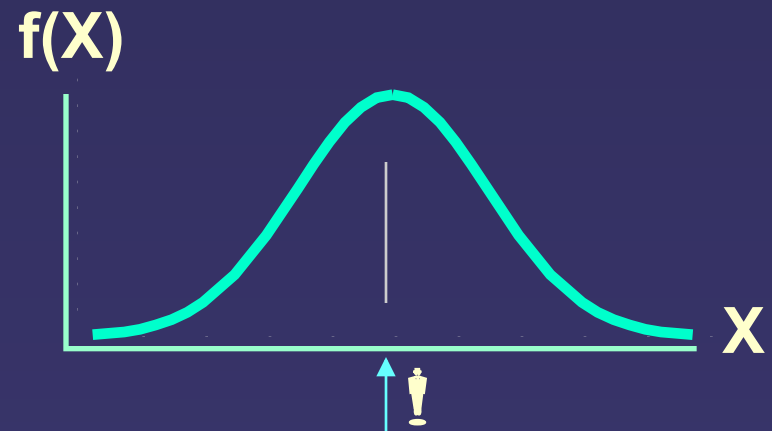
$$F(x) = \frac{x - a}{b - a} \quad \text{if } a \leq x \leq b$$

$$F(x) = 1 \quad \text{if } b < x$$



The Normal Distribution

- ‘Bell Shaped’
- Symmetrical
- Mean, Median and Mode are Equal
- ‘Middle Spread’
Equals 1.33 ?
- Random Variable has
Infinite Range



Mean
Median
Mode

The Mathematical Model

$$f(X) = \frac{1}{\sqrt{2\pi s^2}} e^{-\frac{1}{2s^2}(X - m)^2}$$

$f(X)$ = frequency of random variable X

π = 3.14159; $e = 2.71828$

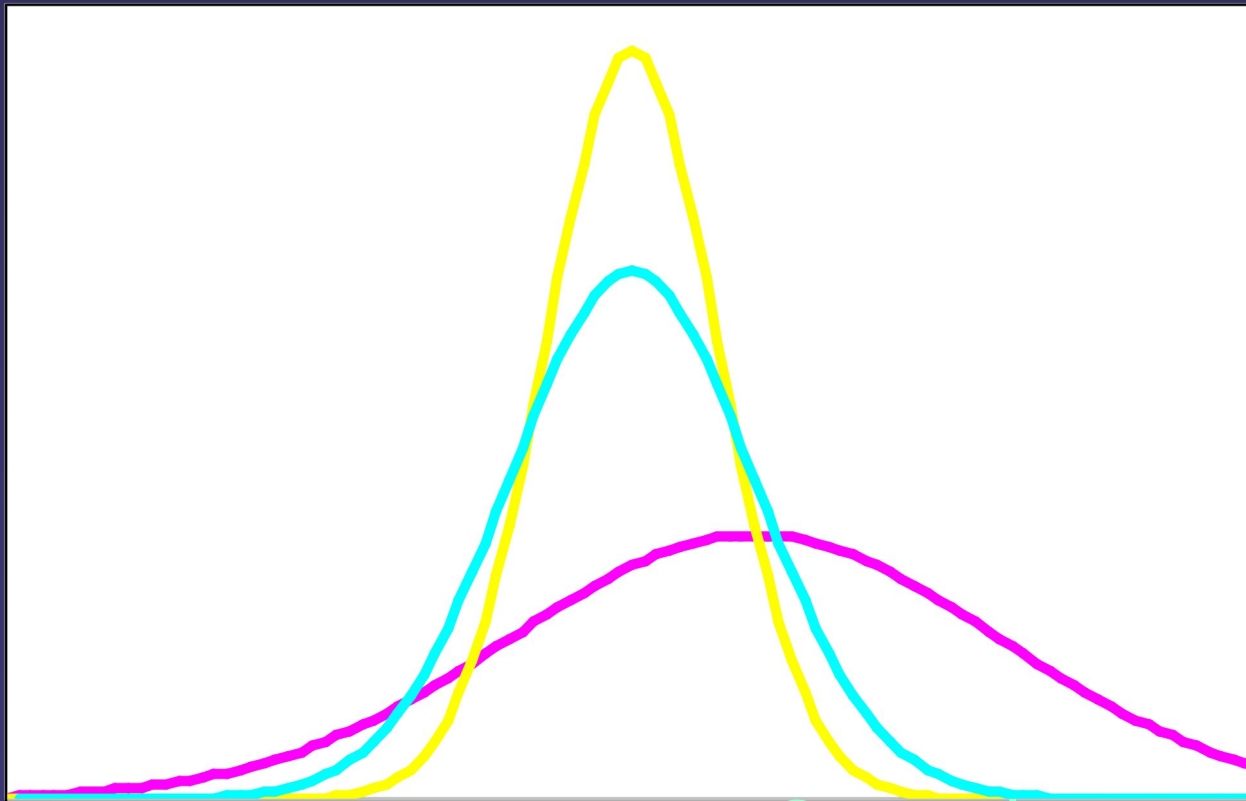
s = population standard deviation

X = value of random variable ($-\infty < X < \infty$)

m

μ = population mean

Many Normal Distributions

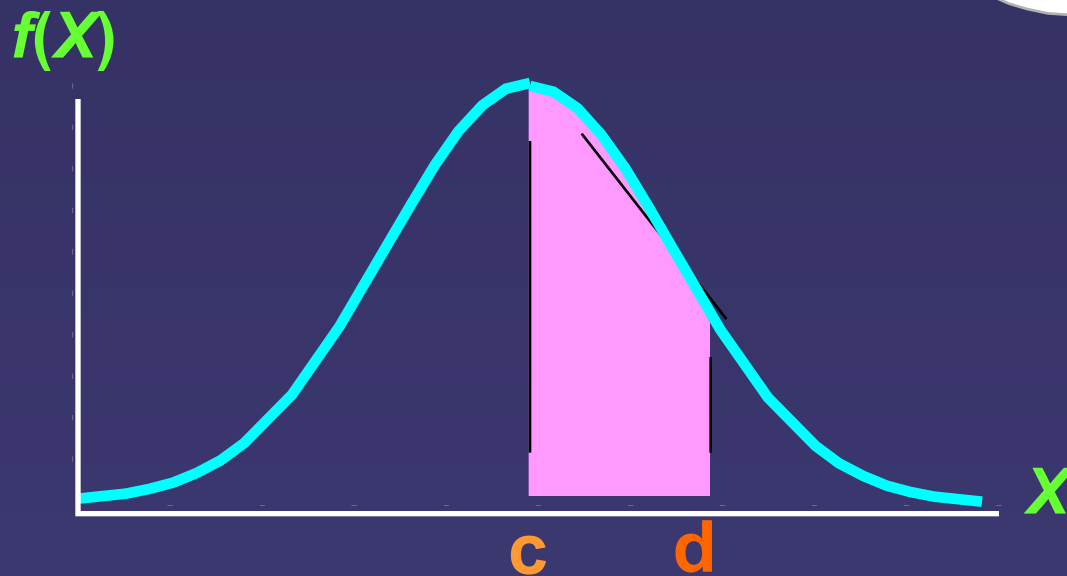


There are
an Infinite
Number

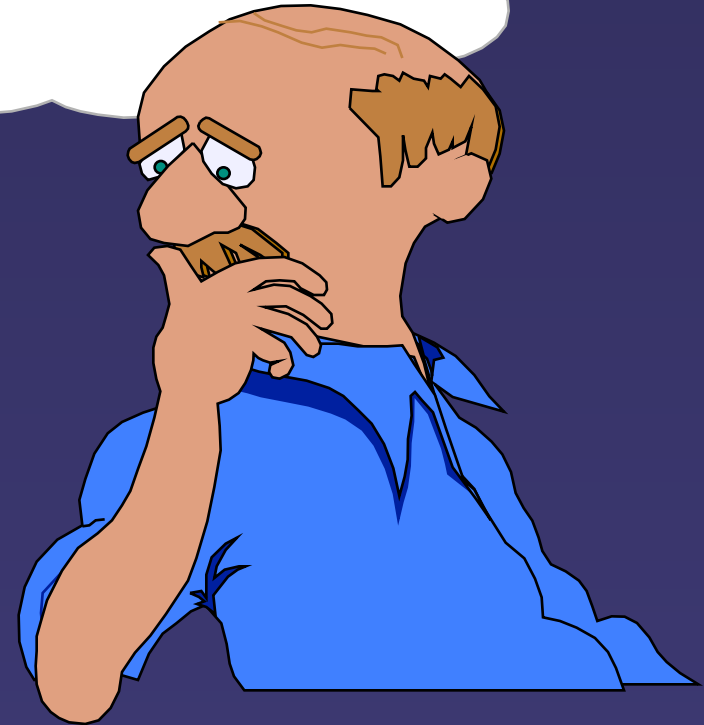
Varying the Parameters μ and σ , we obtain
Different Normal Distributions.

Normal Distribution: Finding Probabilities

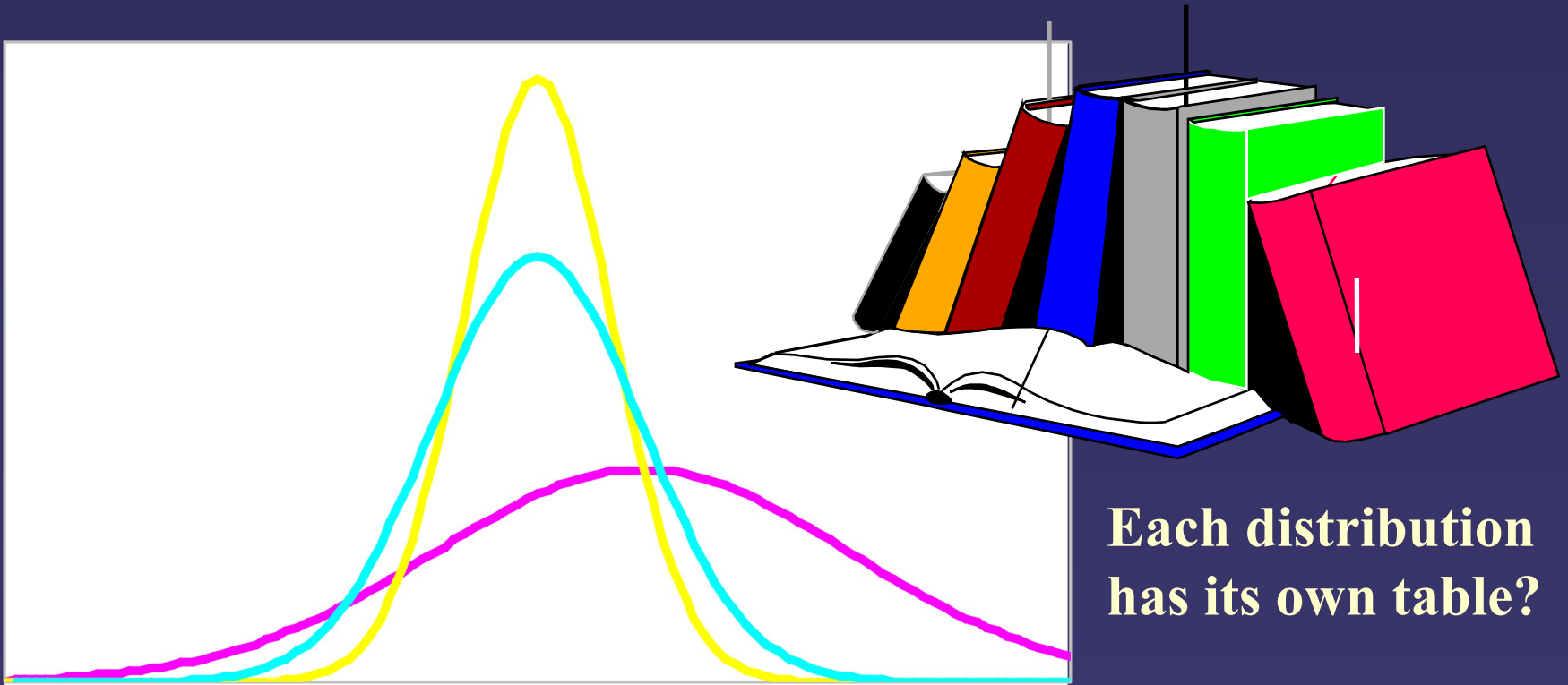
Probability is the
area under the
curve!



$$P(c \leq X \leq d) \bullet ?$$



Which Table?



**Each distribution
has its own table?**

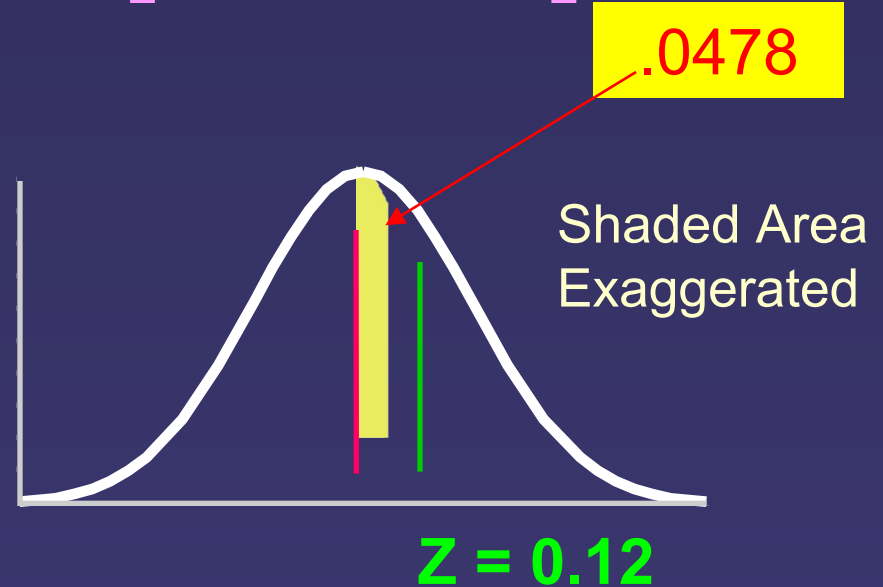
**Infinitely Many Normal Distributions Means
Infinitely Many Tables to Look Up!**

Solution (I): The Standardized Normal Distribution

Standardized Normal Distribution Table (Portion) $\mu = 0$ and $\sigma = 1$

Z	.00	.01	.02
0.0	.0000	.0040	.0080
0.1	.0398	.0438	.0478
0.2	.0793	.0832	.0871
0.3	.0179	.0217	.0255

Probabilities



Only One Table is Needed

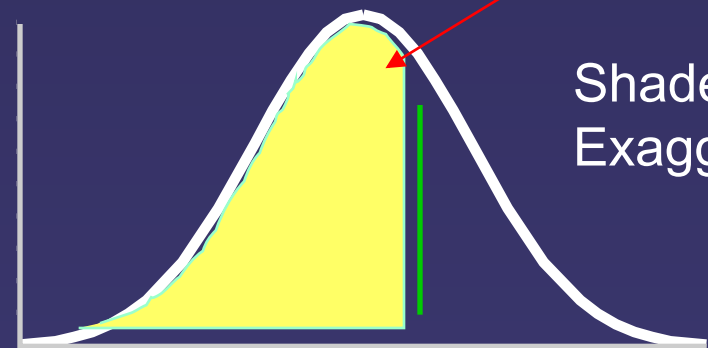
Solution (II): The Cumulative Standardized Normal Distribution

Cumulative Standardized Normal Distribution Table (Portion)

Z	.00	.01	.02
0.0	.5000	.5040	.5080
0.1	.5398	.5438	.5478
0.2	.5793	.5832	.5871
0.3	.5179	.5217	.5255

$m_z = 0$ and $s_z = 1$

.5478



Z = 0.12

Probabilities

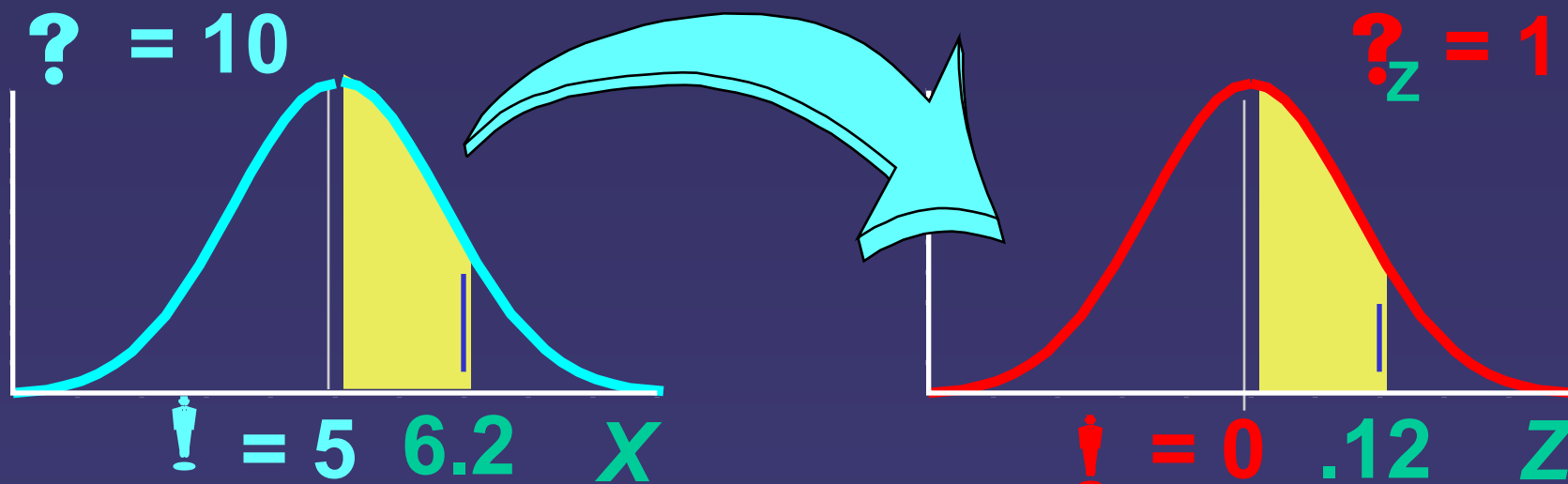
Only One Table is Needed

Standardizing Example

$$Z = \frac{X - m}{s} = \frac{6.2 - 5}{10} = 0.12$$

Normal
Distribution

Standardized
Normal Distribution



Shaded Area Exaggerated

Example:

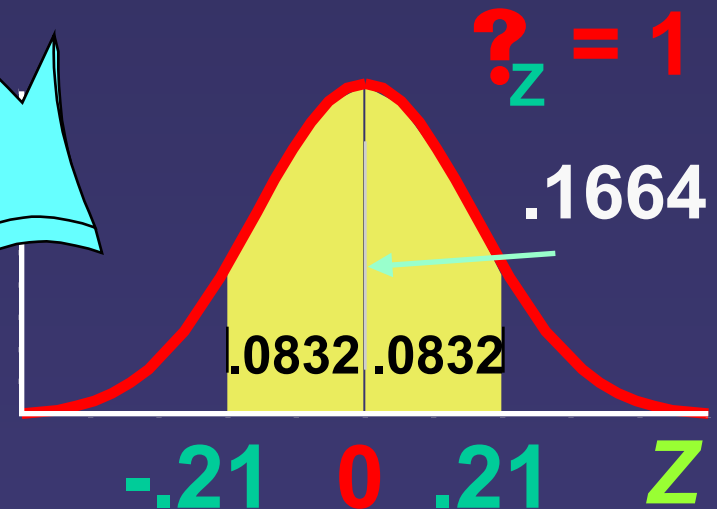
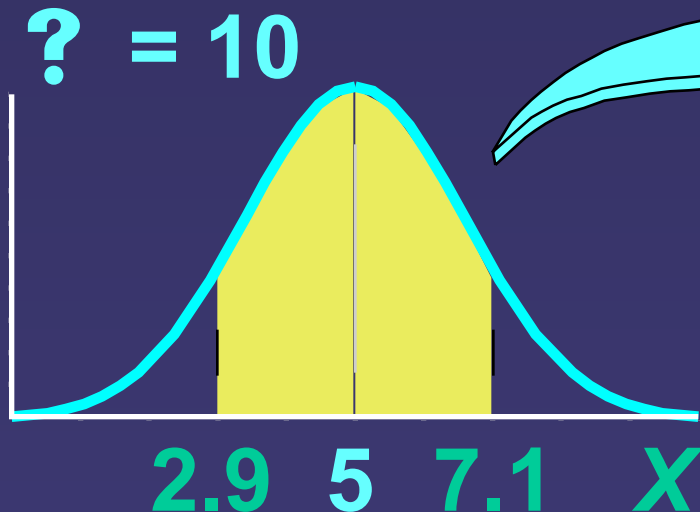
$$P(2.9 < X < 7.1) = .1664$$

$$z = \frac{x - m}{s} = \frac{2.9 - 5}{10} = -.21$$

$$z = \frac{x - m}{s} = \frac{7.1 - 5}{10} = .21$$

Normal
Distribution

Standardized
Normal Distribution



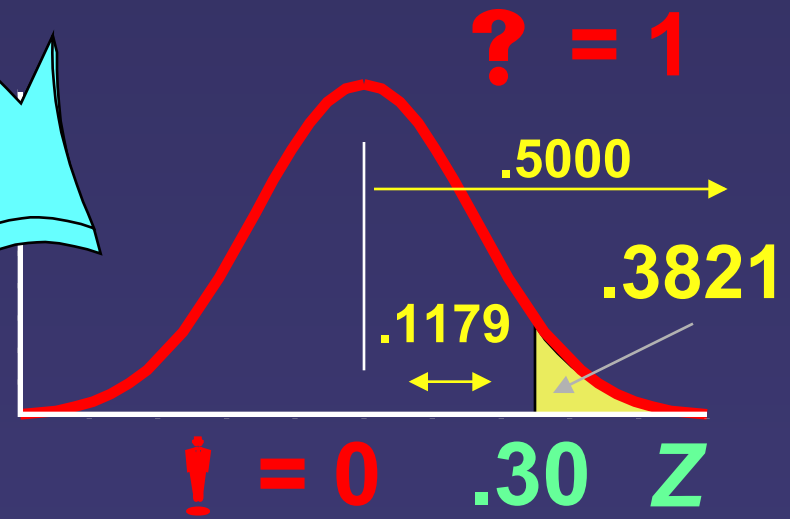
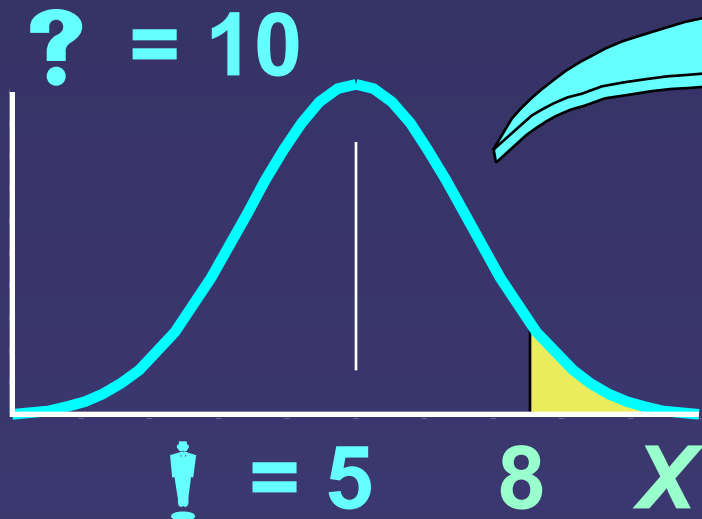
Shaded Area Exaggerated

Example: $P(X \leq 8) = .3821$

$$z = \frac{x - m}{s} = \frac{8 - 5}{10} = .30$$

Normal
Distribution

Standardized
Normal Distribution

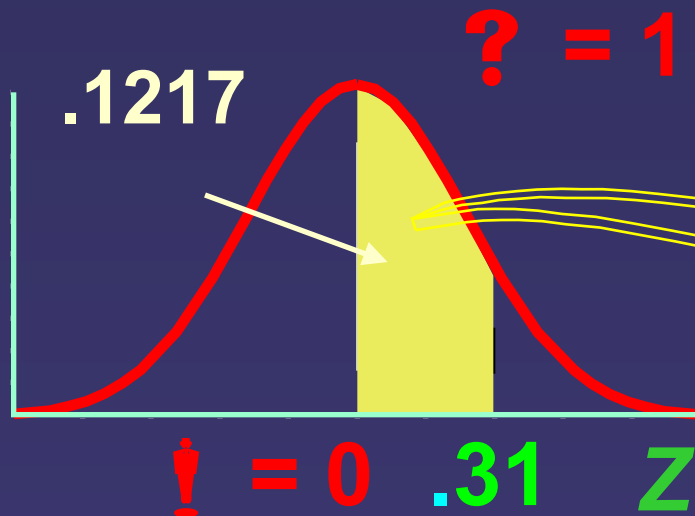


Shaded Area Exaggerated

Finding Z Values for Known Probabilities

What Is Z Given
Probability = 0.1217?

Standardized Normal
Probability Table (Portion)



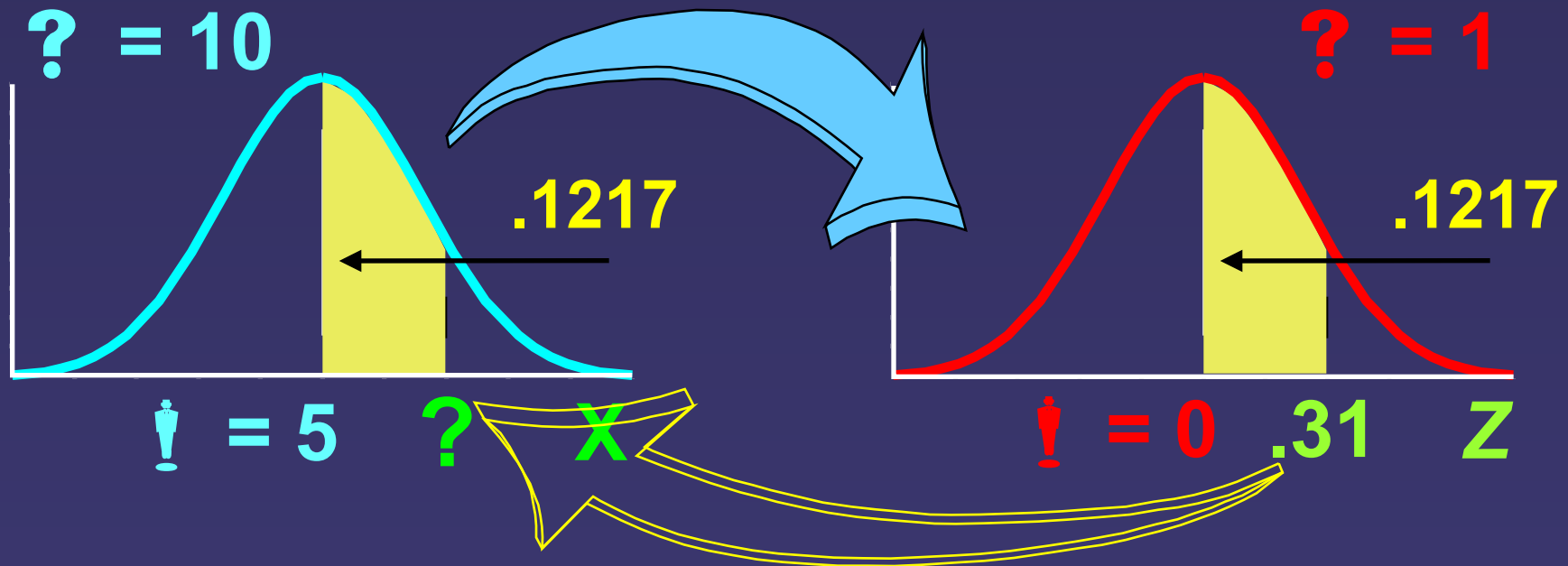
Z	.00	.01	0.2
0.0	.0000	.0040	.0080
0.1	.0398	.0438	.0478
0.2	.0793	.0832	.0871
0.3	.1179	.1217	.1255

Shaded Area
Exaggerated

Recovering X Values for Known Probabilities

Normal Distribution

Standardized Normal Distribution



$$X = \mu + Z \sigma = 5 + 8.1$$

$(0.31)(10) =$ Shaded Area Exaggerated

THANK YOU